Off-shell Supersymmetry and the M-theory Effective Action

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Based on work with
K. Becker, M. Becker, D. Butter, S. Guha, S. Randall, D. Robbins, and A. Sengupta.

Open string effective action

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- Metsaev, Rahkmanov, and Tseytin realize the mistake (R boundary condition) and recalculate with NS to find that the low-energy effective action is Born-Infeld (1987).
- Bagger and Galperin demonstrate that, starting in 4D, $N = 1$ superspace, a second non-linearly-realized supersymmetry uniquely fixes the action to be Born-Infeld (1996). Schematically,

$$F^2 + F^4 + 8 \text{ supercharges} \Rightarrow \text{Born-Infeld}$$

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- Analog of BG result may hold: There are only two supersymmetric densities in flat space. They would be related by $N \geq 2$ supersymmetry. Maybe even fixed by N large enough?

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- Perhaps keeping some $0 < N < 1$ manifest in 11D is enough?

Previous work on fractional superspaces

- Marcus, Sagnotti, and Siegel (1983): 10D SYM in a $\mathbb{R}^{10|4}$. The vector multiplet in this 10D, $N = 1/4$ superspace consists of a real field $V(x, \theta, \bar{\theta}; y)$ familiar from 4D, $N = 1$ and an $SU(3)$ triplet of chiral superfields $\Phi_i(x, \theta; y)$. Action $\sim \int d^{10}x \int d^2\theta W^\alpha W_\alpha +$ Chern-Simons terms.

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- Extended to 11D SG with group at Texas A&M.

<u>Goal:</u> Higher-derivative corrections to the M-theory effective action.

Constructing 11D, $N = 1/8$ Superspace

We construct 11D supergravity in an off-shell superspace $\boldsymbol{M} \sim \boldsymbol{X} \times Y$ with $\boldsymbol{X}$ a 4D, $N = 1$ supermanifold and Y a bosonic 7D manifold.

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1. Decompose 11D SG fields

$$
\begin{array}{ccccc}
e_{\boldsymbol{m}}{}^{\boldsymbol{a}} & \to & e_m{}^a \ , \ \mathcal{A}^i_m \ , \ g_{ij} & \text{with} & a, m = 0, 1, 2, 3 \\
\psi^{\boldsymbol{\alpha}}_{\boldsymbol{m}} & \to & \psi^\alpha_m \ , \ \psi^{\alpha i}_m \ , \ \psi^\alpha_j \ , \ \psi^{\alpha i}_j \ , & \text{and} & i, j = 1, \ldots, 7 \\
C_{\boldsymbol{mnp}} & \to & C_{mnp} \ , \ C_{mn\,i} \ , \ C_{m\,ij} \ , \ C_{ijk} & \text{and} & \alpha = 1, 2
\end{array}
$$

Step 2 (of 6): Reinterpreting the Metric on Y

2 Rewrite the metric g_{ij} on Y in terms of a 3-form (Bryant 1987, Hitchin, Joyce 2000)

$$\sqrt{g}g_{ij} = \frac{1}{144}\epsilon^{k_1\cdots k_7}\varphi_{ik_1k_2}\varphi_{jk_3k_4}\varphi_{k_5k_6k_7}$$

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- If $T_i^j = 0$, the holonomy of ∇ is contained in G_2. The $49 = 1 + 7 + 14 + 27$ components of this tensor make up the torsion forms

$$d\varphi = \tau_0 * \varphi + \tau_1 \wedge \varphi + *\tau_3 \quad \text{and} \quad d(*\varphi) = \tau_1 \wedge *\varphi + \tau_2 \wedge \varphi$$

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- If, furthermore, Y is compact and simply-connected, then $\mathrm{Hol}(\nabla) = G_2$.
- The functional $\int_Y \sqrt{g(\varphi)} = \int_Y \varphi \wedge *\varphi$ is called the Hitchin (2000) functional. When $d\varphi = 0$ its equation of motion is $d(*\varphi) = 0 \Leftrightarrow G_2$ holonomy metrics ("Topological M-theory" Dijkgraaf, Gukov, Neitzke, Vafa 2004).

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- $C_{mn\,i}$ into a chiral spinor $\Sigma_{\alpha i}(x, \theta; y)$ forming a 1-form on Y,
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- $C_{m\,ij}$ into 21 real superfields $V_{ij}(x, \theta, \bar{\theta}; y)$ forming a 2-form on Y,
- $C_{ijk} + i\varphi_{ijk}$ into 35 chiral superfields $\Phi_{ijk}(x, \theta; y)$ forming a 3-form on Y.

All these fields are superconformal primaries of the 4D, $N = 1$ superconformal algebra.

Step 4 (of 6): Chern-Simons Action

4 The last five form a "non-abelian tensor hierarchy" of superfields.[1] Such things have a unique Chern-Simons-like invariant supersymmetrizing the term $\int C \wedge dC \wedge dC$:

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$$-12\kappa^2 \mathcal{L}_{CS} = i \int d^2\theta \mathcal{E} \left[i\Phi_{[0,3]} \wedge \mathbb{G}_{[4,4]} + \Sigma^{\alpha}_{[2,1]} \wedge \mathbb{W}_{[2,6]\alpha} \right] + \int d^4\theta E \left[V_{[1,2]} \wedge \mathbb{H}_{[3,5]} - X_{[3,0]} \mathbb{F}_{[1,7]} \right] + \text{h.c.}$$

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These terms and the others (10 total) are required for gauge invariance.

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4 The last five form a "non-abelian tensor hierarchy" of superfields.[1] Such things have a unique Chern-Simons-like invariant supersymmetrizing the term $\int C \wedge dC \wedge dC$:

$$-12\kappa^2 \mathcal{L}_{CS} = i \int d^2\theta \mathcal{E} \left[i\Phi_{[0,3]} \wedge \mathbb{G}_{[4,4]} + \Sigma^{\alpha}_{[2,1]} \wedge \mathbb{W}_{[2,6]\alpha} \right] + \int d^4\theta E \left[V_{[1,2]} \wedge \mathbb{H}_{[3,5]} - X_{[3,0]} \mathbb{F}_{[1,7]} \right] + \text{h.c.}$$

- $\Phi \wedge \mathbb{G} = G\Phi \wedge d\Phi + \cdots$ contains "G_2 superpotential" (Gukov 1999).
- $\Phi \wedge \mathbb{G} = \frac{1}{2}\Phi \wedge W^{\alpha} \wedge W_{\alpha} + \cdots$ also contains the holomorphic gauge-kinetic function $f(\Phi) = \Phi$ familiar from Seiberg-Witten theory.

These terms and the others (10 total) are required for gauge invariance. Unique invariant containing F-term integrals: $\nexists$ gauge-invariant superpotential or gauge-kinetic function.

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Steps 5 and 6: Complete Action

5 There is a natural lift of the Hitchin functional to 11D, $N = 1/8$ superspace:

$$S_H = -\frac{3}{\kappa^2}\int d^7y \int d^4x \int d^4\theta E(\bar{G}G)^{1/3}\sqrt{g(F)}\, f(x) \quad , \quad x = \frac{H_i g^{ij}(F) H_j}{(\bar{G}G)^{2/3}}$$

Uniquely fixed by local supersymmetry on $\boldsymbol{X}$, diffeomorphism invariance on Y, tensor hierarchy gauge invariance, engineering dimensions, conformal weights, $U(1)_R$ weights, and $GL(7)$ representation.

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The combination $\hat{f} := f - 2xf'$ satisfies the quartic equation

$$\frac{x}{4}\hat{f}^4 + \hat{f}^3 - 1 = 0 \quad \text{and} \quad f(0) = 1 \quad \Rightarrow \quad f(x)$$

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$$V = \tfrac{1}{4} d_X^2 + 2 g_{ij} \boldsymbol{d}^i \boldsymbol{d}^j - d_{\mathbf{7}ij}^2 + \tfrac{1}{2} d_{\mathbf{14}ij}^2 - \tfrac{1}{18} |f_{\mathbf{1}ijk} + \tfrac{i}{2} d_X F_{ijk}|^2 - \tfrac{1}{24} |f_{\mathbf{7}ijk}|^2 + \tfrac{1}{24} |f_{\mathbf{27}ijk}|^2$$

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$$V(x, y) = -\tfrac{21}{16} \tau_0^2 - 15 \tau_1^2 + \tfrac{1}{8} \tau_2^2 + \tfrac{1}{24} \tau_3^2 + \tfrac{7}{4} \sigma_0^2 + 9 \sigma_1^2 + \tfrac{1}{24} \sigma_3^2$$

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$$V(x,y) = -\tfrac{1}{2} R(g) + \tfrac{1}{4 \cdot 4!} F_{ijkl}^2$$

  $=$ Einstein-Hilbert and Maxwell terms on Y from *auxiliary field mechanism*.

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- But the Riemann tensor on Y is real, so S cannot be constructed directly from it. This was also the case for the potential, so perhaps it is generated by some auxiliary field mechanism?
- But the Weyl tensor is in the $\mathbf{70'}$ representation of G_2, and there is no auxiliary field for this!

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   Solution to the $\mathbf{8}$ problem in BI $\overset{?}{\longleftrightarrow}$ solution to $\mathbf{70'}$ problem in M-theory.
   Better yet: If we can show there is no solution to the $\mathbf{8}$ problem, we can safely forget about the $\mathbf{70'}$ problem and go work on something else.

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$$\mathbb{G} = W^{\alpha\beta\gamma} W_{\alpha\beta\gamma}$$

$$\begin{aligned}\mathbb{H}_i = &-\tfrac{1}{4} D^\gamma F_i'^{\alpha\beta} W_{\alpha\beta\gamma} + \tfrac{1}{8} F_i'^{\alpha\beta} \bar{D}^{\dot\alpha} D_\beta G_{\underline{a}} + \tfrac{1}{8} \bar{D}^{\dot\alpha} F_i'^{\alpha\beta} D_\beta G_{\underline{a}} \\ &+ \tfrac{1}{16} \left(D^\beta \bar{D}_{\dot\alpha} F'_{\alpha\beta i} + \tfrac{1}{4} D_\alpha \bar{D}^2 \bar{\lambda}_{\dot\alpha i} \right) G^{\underline{a}} + \tfrac{1}{32} D^2 \lambda_i^\alpha D_\alpha R + \text{h.c.}\end{aligned}$$

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3. This step is trivial in 5D.

Conveniently, this answer can be lifted to 11D, $N = 1/8$ unchanged.

Summary

- We have embedded eleven-dimensional supergravity in an off-shell 11D, $N = 1/8$ superspace.
- We have constructed the linearized off-shell supergeometry for it by starting with the fields and their gauge and local supersymmetry transformations and constructing the Riemann tensor, gravitino curls, and super-4-form field strength.
- We have partially constructed the $(\text{Riemann})^2$ invariants with the intention of using them to construct the leading $\sim R^4$ M-theory corrections.
- Along the way, we have revisited the 5D, $N = 1/2$ analog of this problem and are currently studying the 10D Born-Infeld analog.

Outlook

- Success in the construction of even the most non-trivial part of the $\int C \wedge X_8$ term is not the end of the story:
  1. Chern-Simons-like action is not invariant under all 11D super-diffeomorphisms (cf. 2-derivative action).

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  Outcome expected from general arguments:

  Supersymmetrization of R^4 must be non-polynomial.

Outlook

This would be an 11D supergravity analog of Born-Infeld. If this is correct, it would settle many questions about the definition of the latter but also leave some open and suggest new ones:

- It is not constructed purely from the curvature. Some derivatives thereof will be generated by the superspace integration. These are needed for supersymmetry.
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- Is it the leading low-energy effective action of M-theory? This could be checked by comparing terms available in the literature from amplitude computations.
- Is there a 64-supercharge theory for which the M-theory result is the Goldstone? Witten's 12D theory and E_8 framing (1996)?
- ...

In addition, there are many applications/extensions (superstrings, compactifications, SFT, ...).

THANK YOU!

NATH fields and transformations

$$
\begin{aligned}
\delta_0 \Psi_{\alpha i} &= \Xi_{\alpha i} + \bar{G}\, \mathcal{G}_{ij} \nabla_\alpha \Omega^j + i\, W_{\alpha ij}\, \bar{\Omega}^j \ , \\
\Delta_1 \Phi &= \Xi^\alpha \wedge W_\alpha + \bar{\nabla}^2 \Big(\frac{i}{2} \iota_{\bar{\Omega}} \mathcal{U} + \frac{1}{4} \iota_{\bar{\Omega}} (F \wedge H) \Big) \ , \\
\Delta_1 V &= -\frac{i}{2} \bar{G} \iota_\Omega F + \frac{i}{2} G \iota_{\bar{\Omega}} F \ , \\
\Delta_1 \Sigma_\alpha &= -G\, \Xi_\alpha \ , \\
\Delta_1 X &= -\frac{i}{2} \bar{G}\, \iota_\Omega H + \frac{i}{2} G\, \iota_{\bar{\Omega}} H \ , \\
\delta_1 \mathcal{V} &= -\frac{1}{2} \bar{G}\, \Omega - \frac{1}{2} G\, \bar{\Omega} \ .
\end{aligned}
$$

$$
\mathcal{U}_{ijkl} := 3\, \epsilon_{ijklmnp} \frac{\partial}{\partial F_{mnp}} \Big(\sqrt{g}\, f\, (G\bar{G})^{1/3} \Big)
$$

$$
\mathcal{G}_{ij} = (G\bar{G})^{-1/3} \Big(\hat{f}^{-1} g_{ij} + \frac{i}{2} (G\bar{G})^{-1/3} \hat{f} F_{ijk}\, g^{kl}\, H_l \Big)
$$

$$
\begin{aligned}
\delta_0 \Psi_{\alpha i} &= \Xi_{\alpha i} + g_{ij} D_\alpha \Omega^j \ , \\
\delta_1 \Phi_{ijk} &= \tfrac{1}{2i} \tilde{\varphi}_{ijkl} \bar{D}^2 \bar{\Omega}^l \ , \\
\delta_1 V_{ij} &= \tfrac{1}{2i} \varphi_{ijk} (\Omega^k - \bar{\Omega}^k) \ , \\
\delta_1 \Sigma_{\alpha i} &= -\Xi_{\alpha i} \ , \\
\delta_1 \mathcal{V}^i &= -\tfrac{1}{2} (\Omega^i + \bar{\Omega}^i) \ .
\end{aligned}
$$

NATH field strengths and Bianchi identities

$$
\begin{aligned}
E &= \partial\Phi \\
F &= \tfrac{1}{2i}\left(\Phi - \bar{\Phi}\right) - \partial V \\
W_\alpha &= -\tfrac{1}{4}\bar{\nabla}^2\nabla_\alpha V + \partial\Sigma_\alpha + \iota_{\mathcal{W}_\alpha}\Phi \\
H &= \tfrac{1}{2i}\left(\nabla^\alpha\Sigma_\alpha - \bar{\nabla}_{\dot{\alpha}}\bar{\Sigma}^{\dot{\alpha}}\right) - \partial X - \omega_{\mathsf{h}}(\mathcal{W}, V) \\
G &= -\tfrac{1}{4}\bar{\nabla}^2 X + \iota_{\mathcal{W}^\alpha}\Sigma_\alpha
\end{aligned}
$$

$$
\begin{aligned}
0 &= -\partial E \\
\tfrac{1}{2i}\left(E - \bar{E}\right) &= \partial F \\
-\tfrac{1}{4}\bar{\nabla}^2\nabla_\alpha F &= -\partial W_\alpha - \iota_{\mathcal{W}_\alpha} E \\
\tfrac{1}{2i}\left(\nabla^\alpha W_\alpha - \bar{\nabla}_{\dot{\alpha}}\bar{W}^{\dot{\alpha}}\right) &= \partial H + \omega_{\mathsf{h}}(\mathcal{W}, F) \\
-\tfrac{1}{4}\bar{\nabla}^2 H &= -\partial G - \iota_{\mathcal{W}^\alpha} W_\alpha \\
\bar{\nabla}_{\dot{\alpha}} G &= 0
\end{aligned}
$$

$$
\omega_{\mathsf{h}}(\chi_\alpha, v) := \iota_{\chi^\alpha}\nabla_\alpha v + \iota_{\bar{\chi}_{\dot{\alpha}}}\bar{\nabla}^{\dot{\alpha}} v + \tfrac{1}{2}\left(\iota_{\nabla^\alpha\chi_\alpha} v + \iota_{\bar{\nabla}_{\dot{\alpha}}\bar{\chi}^{\dot{\alpha}}} v\right)
$$

Chern-Simons-like invariant

$$
\begin{aligned}
-12L_{CS} = i \int d^2\theta\, \mathcal{E}\, \Phi \left[EG + \tfrac{1}{2} W^\alpha W_\alpha - \tfrac{i}{4} \bar{\nabla}^2 (FH) \right] \\
+ \int d^4\theta\, E\, V \left[\left(E + \bar{E} \right) H + \omega(W, F) - i \nabla^\alpha F \iota_{\mathcal{W}_\alpha} F + i \bar{\nabla}_{\dot{\alpha}} F \iota_{\bar{\mathcal{W}}^{\dot{\alpha}}} F \right] \\
+ i \int d^2\theta\, \mathcal{E}\, \Sigma^\alpha \left[E W_\alpha - \tfrac{i}{4} \bar{\nabla}^2 (F \nabla_\alpha F) \right] \\
- \int d^4\theta\, E\, X \left[\left(E + \bar{E} \right) F \right] + \text{h.c.}
\end{aligned}
$$